

4(a)	the number of wavefronts/crests/troughs passing a fixed point per unit time or the number of oscillations per unit time (of source / point on wave / particle of medium)	B1
4(b)(i)	$T = 4 \times 0.50 \times 10^{-3}$ $(= 2.0 \times 10^{-3} \text{ s})$	C1
	$f = 1 / 2.0 \times 10^{-3}$ $= 500 \text{ Hz}$	A1
4(b)(ii)	amplitude $= 2.8 \times 0.20$ $= 0.56 \text{ V}$	A1
4(c)	period same as original trace	B1
	sinusoidal wave of constant amplitude less than 2.8 cm throughout	M1
	amplitude 1.4 cm	A1

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Question	Answer	Marks
4(d)(i)	when (two or more) waves <u>meet</u> (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(d)(ii)	node-to-node separation is $\lambda / 2$ or microphone moves through 3 node-to-node separations or $d = 1.5\lambda$	C1
	$\lambda = 1.05 / 1.5$ $= 0.70 \text{ m}$	A1
4(d)(iii)	$v = f \times \lambda$	C1
	$= 500 \times 0.70$ $= 350 \text{ m s}^{-1}$	A1